

PROBLEM & SOLUTION: BUS TRIP ANALYSIS

Chapter 14 | Description of Motion

Problem Statement

Exercise Context

A bus trip consists of 3 phases:

- **Phase 1:** Bus moving at $V_1 = 70 \text{ km/h}$ for $\Delta t_1 = 45 \text{ min}$.
- **Phase 2:** Bus *stopped* to refuel for $\Delta t_2 = 0.25 \text{ h}$.
- **Phase 3:** Bus continued moving at $V_3 = 100 \text{ km/h}$ for a distance $d_3 = 80 \text{ km}$.

Questions:

- Calculate the trip total duration, Δt_{tot} .
- Calculate the trip total distance, d_{tot} .
- Deduce the bus average speed for the whole trip, V_{av} .

Step-by-Step Solution

2.1 1. Preliminary Conversions and Data Preparation

To solve this problem, we must ensure all time values are in hours (h) and all distances are in kilometers (km).

Analysis of Phase 1:

- Time: $\Delta t_1 = 45 \text{ min} = \frac{45}{60} \text{ h} = \mathbf{0.75 \text{ h}}$.
- Distance: $d_1 = V_1 \times \Delta t_1 = 70 \text{ km/h} \times 0.75 \text{ h} = \mathbf{52.5 \text{ km}}$.

Analysis of Phase 2:

- The bus is stopped, so distance $d_2 = \mathbf{0 \text{ km}}$.
- Duration is given as $\Delta t_2 = \mathbf{0.25 \text{ h}}$.

Analysis of Phase 3:

- Distance is given as $d_3 = \mathbf{80 \text{ km}}$.
- Time: $\Delta t_3 = \frac{d_3}{V_3} = \frac{80 \text{ km}}{100 \text{ km/h}} = \mathbf{0.8 \text{ h}}$.

2.2 2. Calculations

a) Total Duration (Δt_{tot}):

$$\Delta t_{tot} = \Delta t_1 + \Delta t_2 + \Delta t_3$$

$$\Delta t_{tot} = 0.75 + 0.25 + 0.8 = \mathbf{1.8 \text{ h}}$$

b) Total Distance (d_{tot}):

$$d_{tot} = d_1 + d_2 + d_3$$

$$d_{tot} = 52.5 + 0 + 80 = \mathbf{132.5 \text{ km}}$$

c) Average Speed (V_{av}):

$$V_{av} = \frac{d_{tot}}{\Delta t_{tot}}$$

$$V_{av} = \frac{132.5 \text{ km}}{1.8 \text{ h}} \approx \mathbf{73.61 \text{ km/h}}$$

Teacher Tip: Common Pitfall

Remind students that "Average Speed" is **not** the average of the speeds $(V_1 + V_3)/2$. It must always be calculated using the total distance divided by the total time, including stop durations where $V = 0$.